	$= 5.27 \times 10^{-4}$ A $= 0.53$ mA	
29	**C** Option A: can be drawn from the observation that most of the α-particles go straight through or are deflected by a small angle. Option B: can be drawn from the observation that a very small proportion are deflected by more than 90°, effectively rebounding towards the source of the α-particles, so the nucleus must be massive to cause the rebounding of the α-particles. Option C: the deflection and rebounding of α-particles can show that the nucleus is positively charged, but it cannot show that there are neutrons in the nucleus. At the time of the experiment, the neutron has not yet been discovered. Option D: can be drawn from the observation that there are α-particles which are deflected or rebounded.	**E**
30	**A**	**A**